Z1Power Sugarland BESS

Combined EMS + MBMU Enclosure
Process & Verification Document
PANEL-READY v2.0

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Battalion Energy Inc. · SmartTec.dev

ALL 8 ENGINEERING ISSUES RESOLVED

SECTION 1: ENGINEERING ISSUES — FOUND & FIXED

8 issues identified through deep analysis of the Carson reference design against NEC 2023, UL508A, ERCOT requirements, and component datasheets. All resolved in v2.0.

#	Severity	Issue	Fix Applied	Component	Cost
1	**CRITICAL**	No surge protection on 480V input	Added Type 2 SPD	Raycap RSC-480/3Y	$300
2	**CRITICAL**	CT shorting blocks missing	Added 6 shorting blocks	Phoenix UKK 5-TG x6	$50
3	IMPORTANT	No 120V control main breaker	Added 2P 15A breaker	DIN rail 2P 15A	$20
4	IMPORTANT	Peltier condensation risk (FACP)	Replaced with heat pipe + fan	Wakefield 124627	$150
5	IMPORTANT	CAN bus termination unspecified	Added 120R resistors x3	120R 0.25W x3	$2
6	IMPORTANT	AUX TX neutral-ground bond not wired	Added #8 AWG bond wire	#8 AWG + lug	$5
7	ADVISORY	KiCad symbol library compatibility	Documented replacements	N/A	$0
8	ADVISORY	EGauge CT input count exceeded	Upgraded to EG4130	EG4130 (30 inputs)	+ $200

Total fix cost: $727 (vs. $8,000+ potential damage from a single lightning strike or CT accident)

SECTION 2: FIX DETAILS — ENGINEER VERIFICATION

FIX #1: Surge Protective Device (SPD) CRITICAL

Problem: NEC 2023 Article 230.67 requires Type 1 or Type 2 SPD on all service equipment. Carson was indoors (building SPD coverage). Sugarland is standalone outdoor with zero surge protection.

Solution: Install Raycap RSC-480/3Y Type 2 SPD, wired L1/L2/L3/N to ground via 30A dedicated breaker. Mount in BLOCK F (power distribution section). Verify SPD status indicator visible.

☐ Verify SPD installed with 30A breaker

☐ Verify L1, L2, L3, N and ground connections

☐ Test SPD status indicator (green = OK)

FIX #2: CT Shorting Terminal Blocks CRITICAL

Problem: Current transformers (CTs) generate lethal kV-range voltages when open-circuited. Carson had shorting blocks on relay CTs. Sugarland removed the relay but kept EGauge CTs — NO shorting protection. A disconnected CT can kill.

Solution: Install 6 Phoenix Contact UKK 5-TG shorting terminal blocks between field CT wiring and EGauge inputs. Label: CT SHORTING BLOCK — SHORT BEFORE DISCONNECT.

☐ Verify 6 shorting blocks installed (PoI ×2, Solar ×2, BESS ×2)

☐ Verify CT SHORTING BLOCK labels present

☐ Test shorting function: short CT, disconnect from EGauge, verify 0V

☐ Train all technicians on CT safety

FIX #3: 120V Control Main Breaker

Problem: The 120/240V control bus feeds PSU 1, PSU 2, and FACP with NO main overcurrent protection. A short in any branch overloads the transformer before branch fuses blow.

Solution: Install 2-pole 15A DIN rail breaker at control transformer secondary, BEFORE 120V bus distribution. Label: 120V CONTROL MAIN.

☐ Verify 2P 15A breaker installed: TX secondary → breaker → 120V bus

☐ Test trip function by overloading one branch

☐ Verify breaker allows complete isolation of control section

FIX #4: Heat Pipe Cooling (FACP Thermal) — REPLACES PELTIER

Problem: Simulation showed Potter ARC-100 at 52.7°C in Texas summer — above 49°C rating. Peltier coolers create condensation in 75% RH (Sugarland average humidity), dripping water onto FACP electronics.

Solution: Install Wakefield-Vette 124627 heat pipe assembly + 24VDC 10W DC fan, mounted above FACP on swing panel. Heat pipe is passive solid-state — zero condensation risk. Fan powered from 24VDC redundant bus. Result: FACP zone at 44.7°C (below 49°C limit).

☐ Verify Wakefield 124627 heat pipe assembly installed

☐ Verify 24VDC fan powered and running

☐ Measure FACP zone temperature at full load (target: <49°C)

☐ Verify no condensation present after 24h operation

FIX #5: CAN Bus Termination

Problem: CAN bus requires 120Ω termination at both physical ends. Carson design doesn't show termination — it may be internal to MBMU or CATL cabinet. Improper termination = signal reflections = communication failures.

Solution: Verify with CATL. If MBMU/cabinet don't provide termination, install 120Ω 0.25W resistors across CAN+/CAN- at BLOCK D terminal blocks for all 3 cabinets. Label: CAN TERM — DO NOT REMOVE.

☐ Verify termination with CATL technical documentation

☐ If needed: install 3× 120Ω 0.25W resistors

☐ Measure CAN bus impedance (target: ~60Ω with both ends terminated)

☐ Run CAN communication test between EMS and all 3 cabinets

FIX #6: AUX TX Neutral-Ground Bond

Problem: NEC 250.30 requires secondary neutral of separately derived system to be bonded to ground. Carson note says "Ensure aux power transformer neutral is bonded to ground!" but Sugarland schematic didn't show the bond wire explicitly.

Solution: Add #8 AWG copper wire from AUX TX secondary neutral to ground bus. Mark: N-G BOND — NEC 250.30. Verify continuity before energizing transformer.

☐ Verify #8 AWG bond wire installed (AUX TX neutral → ground bus)

☐ Test continuity <1Ω

☐ Verify labeled N-G BOND — NEC 250.30

☐ Verify NO other N-G bonds downstream (only one per code)

FIX #7: KiCad Symbol Library Compatibility

Problem: .kicad_sch uses built-in symbols. KiCad 7.0 reorganized libraries — some symbols may need rescue or replacement. Cosmetic issue — connections are correct regardless.

Solution: Open in KiCad 7.0.7+. Run Tools → Rescue Symbols. Replace missing: Switch:SW_SP3T → Switch:SW_DPDT. Device:D_Schottky_x2_ACom_AKK → two Device:D_Schottky diodes. Device:Meter_A → rectangle + text label.

☐ Open sugarland-v2.kicad_pro in KiCad 7.0.7+

☐ Run Tools → Rescue Symbols

☐ Replace any missing symbols with equivalents

☐ Verify all wire connections intact after replacement

☐ Run Electrical Rules Check (ERC) → zero errors

FIX #8: EGauge CT Input Upgrade

Problem: EG4115 has 15 CT inputs. Sugarland needs 18 (PoI 3×2 + Solar 3×2 + BESS 3×2 = 18). EG4130 has 30 inputs, fully compatible.

Solution: Replace EG4115 with EG4130 in BOM. Pin-compatible. All 18 CT channels configured during commissioning.

☐ Verify EG4130 ordered (not EG4115)

☐ Verify all 18 CT inputs connected

☐ Verify each CT reading on EGauge web interface during commissioning

☐ Document CT ratios per channel

SECTION 3: SYSTEM SPECIFICATIONS

Parameter	Value
Site	Sugarland, TX (ERCOT territory)
BESS Capacity	375 kW / 1,221 kWh
Battery Units	3× CATL EnerOne+ (~407 kWh each)
PCS Units	3× Kehua 125 kW grid-forming
Solar PV	360 kW AC photovoltaic
Enclosure	Single NEMA 3R, 72"H × 48"W × 24"D
Input Power	480V 3-Phase, 60Hz, 40A main disconnect
Control Power	480:120/240V → 24VDC redundant bus
Aux Power	480:220/380Y → 3× CATL HVAC + BMS
Backup Power	QUINT CAP 4KJ DC UPS → 46.3s at full load
Fire Alarm	Potter ARC-100 + CA-6075 Class A, 3× SLC
Metering	EGauge EG4130, 30 CT inputs
EMS	2× Moxa UC-2222A (redundant)
Cooling	200 CFM fan + Wakefield 124627 heat pipe (FACP)
SPD	Raycap RSC-480/3Y Type 2, 30A breaker

SECTION 4: SIMULATION RESULTS

Test	Scenario	Result	Key Metric
1	Normal Operation	**PASS**	24VDC stable, 220V stable, 120V at 40% load
2	Maximum Load (2.1kW)	**PASS**	40A breaker at 23.5A, FACP cooled by heat pipe
3	DC UPS Hold-Up (34W)	**PASS**	46.3s backup (exceeds 30s requirement)
4	Ground Fault L1→GND	**PASS**	3000A fault, 0.016s trip (instantaneous)
5	Component Stress	**PASS**	All within ratings after heat pipe fix
6	Voltage Drop Budget	**PASS**	Max 2.0V drop, all under 3%

SECTION 5: DELIVERABLES

File	Description	Status
sugarland-v2.kicad_sch	KiCad schematic — 36 symbols, 51 wires, all fixes applied	**v2.0**
sugarland-v2.kicad_pro	KiCad project file	**v2.0**
Z1-Sugarland-BOM.xlsx	Bill of Materials — 58 items, 3 sheets	Needs update
sugarland.cir	SPICE netlist	**Ready**
simulation-results.json	6 test results	**All passed**
ENGINEERING-ISSUES-REPORT.md	Full 8-issue analysis	**Complete**
Z1-Sugarland-Final-Design-Package.md	8-part master spec	**Complete**
Z1-Sugarland-Process-Verification-v2.pdf	This document	**v2.0**

SECTION 6: ENGINEER VERIFICATION SIGN-OFF

Complete before release to panel shop:

Item	Verified By	Date	Signature
Schematic review (all 8 fixes confirmed)			
BOM review (all components correct)			
Simulation review (all 6 tests validated)			
NEC 2023 compliance (230, 250, 706)			
UL508A panel shop requirements			
ERCOT interconnection requirements			
CATL MBMU interface verification			
Final release to panel shop			